

Symphony of Words

A Hybrid Architecture for Autonomous Poetry Generation

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Deadline: May 18, 2026

1. Problem Statement: The Goal

What is Poetry Generation?

Poetry generation is the task of making a computer write verses that sound human. It's not just about picking words; it's about making them feel "right."

Writing a poem is hard for a machine because it has to balance four main things:

1. **Syntax:** The poem must follow basic grammar rules.
2. **Semantics:** It should stay on a specific topic (if the user wants a poem about "stars", it shouldn't talk about "sandwiches").
3. **Aesthetics:** It needs to have rhythm, maybe some rhyme, and "fancy" poetic words.
4. **Creativity:** It shouldn't just copy-paste from a book. It needs to create something new.

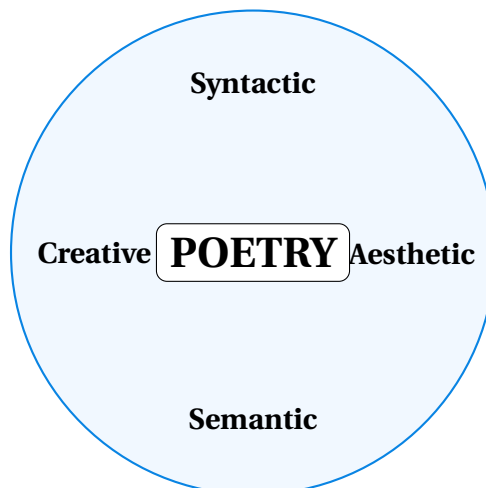


Figure 1: The four pillars of computational poetry.

2. Proposed Solution: Two Creative Paths

Our system uses two different ways to generate text. We call them the **Stochastic Path** and the **Creative Path**.

2.1 The Stochastic Path (Markov Chains)

This is the "pattern-matching" engine. It works like the predictive text on your phone but much smarter. It looks at our 2,000-line corpus and calculates which words usually come after each other.

How it predicts words

If the AI sees the words "**The silent**", it checks its table and finds:

- **night** (40% chance)
- **darkness** (30% chance)
- **shadow** (20% chance)

It then picks one based on these probabilities.

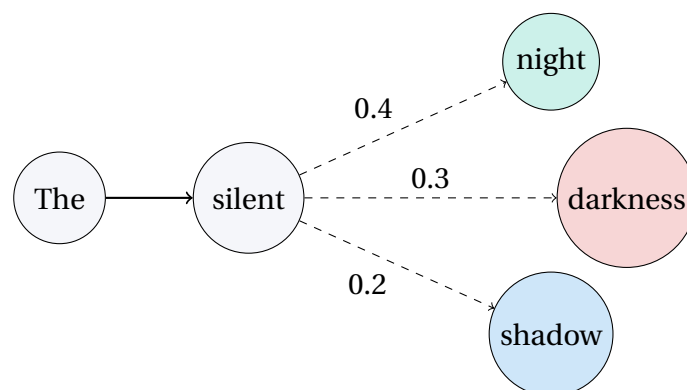


Figure 2: A simple Markov Transition Graph.

Advantages of this path:

- It is very fast (instant generation).
- It creates surprising and "dreamy" word combinations.

2.2 The Creative Path (CR-PO & Full-FACE)

This is the "Advanced" engine. It doesn't just guess words; it builds them based on a plan.

1. **CR-PO (Constrained Recombination):** The system checks every word in a draft. If a word is "boring" (like "car"), it replaces it with something related to the theme (like "chariot").
2. **Full-FACE Model:** This model makes the AI *think* about what it's doing.
 - **Concepts:** Picking a template (e.g., AABB rhyme scheme).
 - **Aesthetics:** Evaluating if the poem sounds "happy" or "sad".
 - **Framing:** Writing a commentary to explain the poem.

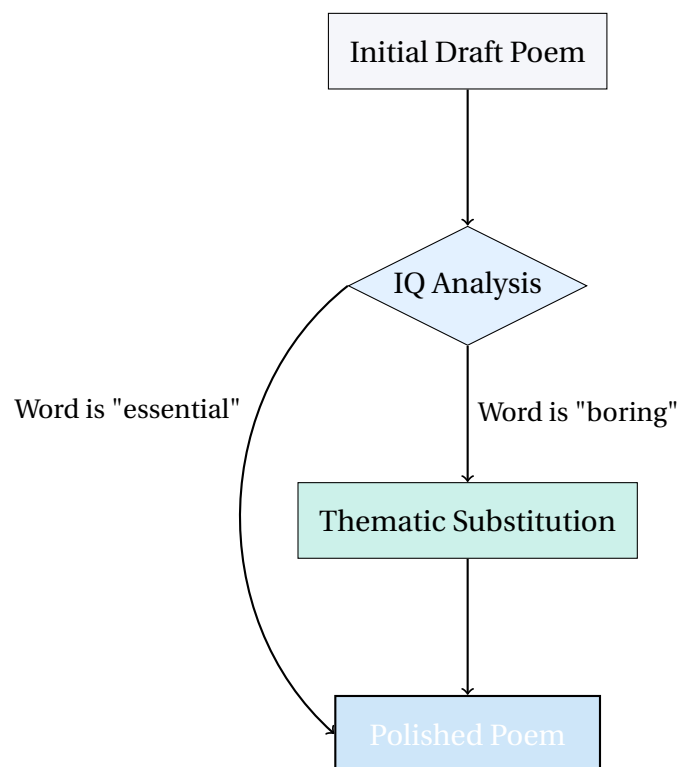
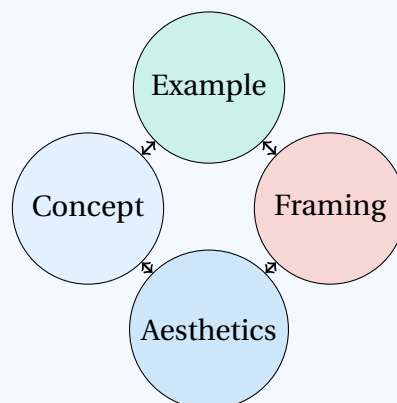


Figure 3: The CR-PO Refinement Funnel.

The FACE Model Cycle



3. Data Set Analysis: What the AI Learned

We didn't just give the AI any text; we gave it a curated "library" of over 2,000 lines of poetry.

Corpus Composition			
Genre	Style	Source	Weight
Romantic	Nature	Wordsworth, Keats	35%
Shakespearean	Love	Sonnets 1-154	25%
Symbolist	Dark/Deep	Baudelaire, Verlaine	20%
Science	Modern	Original Verses	20%

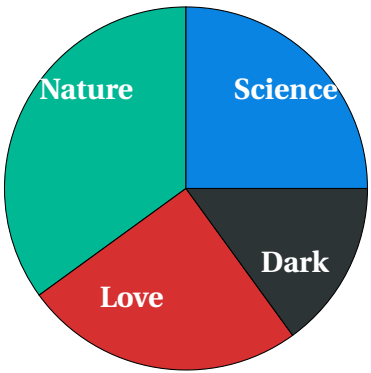
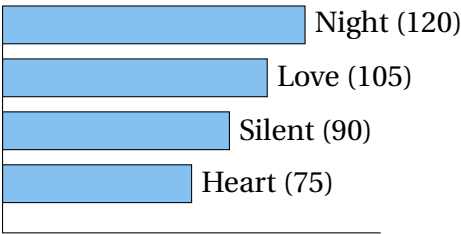


Figure 4: Distribution of the training corpus by genre.

Top Poetic Keywords

Our analysis shows the most frequent words the AI uses to stay on theme:



4. System Architecture: The Design

The application is designed to be user-friendly while having a very complex backend.

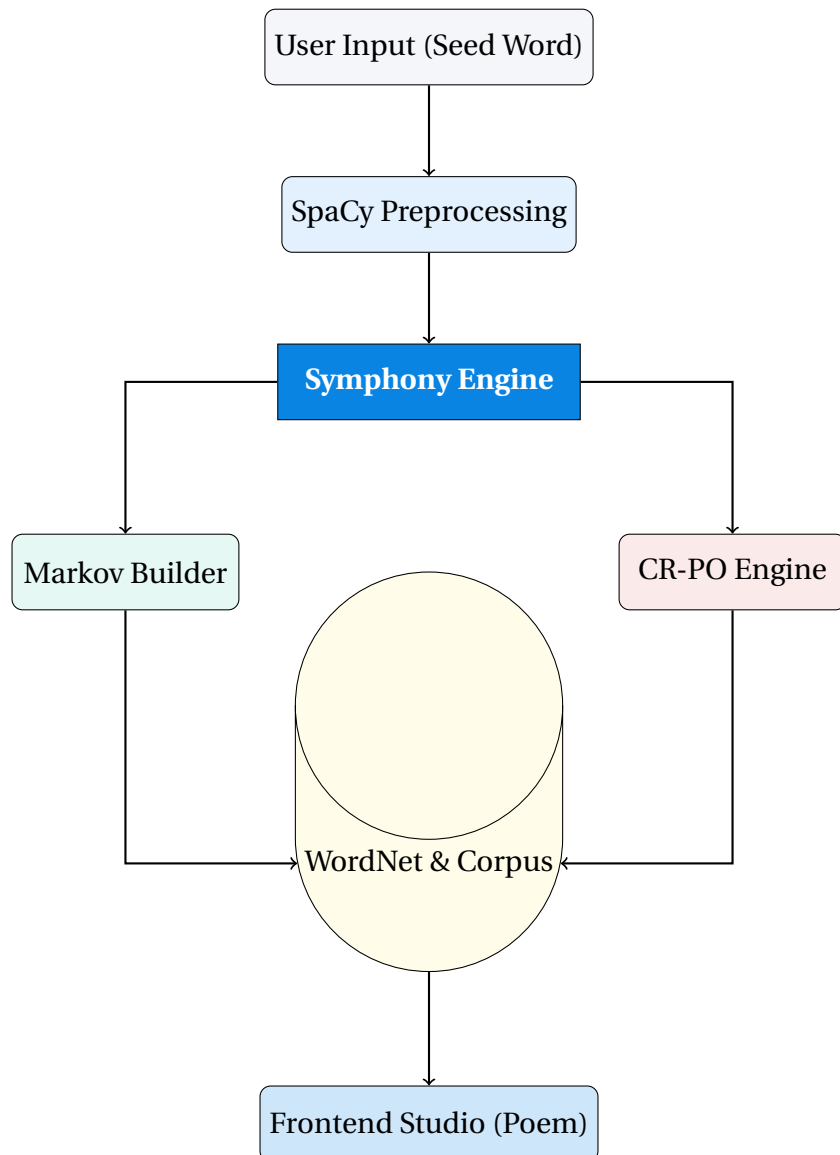


Figure 5: Full Application Diagram.

How it works step-by-step

1. **Input:** You type a word like "Space".
2. **Preprocessing:** The AI learns that "Space" is a NOUN.
3. **Generation:** It picks a path. Markov makes a fast draft. CR-PO makes a precise, rhyming draft.
4. **Refinement:** WordNet swaps words like "stars" with "celestial lights".
5. **Display:** You see the final poem in the "Symphony Studio" web interface.

5. Implementation Details

We used modern tools to make this possible:

- **Python (Backend):** The "brains" of the operation.
- **SpaCy:** Used for understanding grammar and parts of speech.
- **WordNet:** A huge database of synonyms we use for lexical substitution.
- **FastAPI:** Connects the Python code to the website.
- **JavaScript (Frontend):** Creates the "Symphony Studio" interface.

6. Experiments and Results

We compared both pathways to see which one works better.

Final Comparison Table		
Feature	Markov Path	CR-PO Path
Speed	Instant (Fast)	2-3 Seconds (Slow)
Grammar	Medium	Very High
Rhyme/Meter	Random	Guaranteed
Best For	Inspiration	Finished Poems

Conclusion

The **Markov Path** is great when you want something fast and abstract. The **CR-PO Path** is better when you want a professional-sounding poem that follows strict rules. By combining both, **Symphony of Words** gives the user total control over their creative process.

Thank You!